

NeuroAI as a Synthetic Cognitive Laboratory

NeuroAI as a Synthetic Cognitive Laboratory

Connectome Runtime, Controlled Perturbation, Battle Ops, And Brain-Powered Benchmarking

Author: Alfredo Medina Hernandez

Affiliation: Medina Research

Date: June 12, 2026

Authority state: Public-safe Zenodo release candidate from build-derived NeuroAI substrate

Release boundary: Synthetic cognition and simulation architecture only; no clinical, therapeutic, biological-consciousness, or deployed operational claim

Abstract

NeuroAI is a build-derived synthetic cognition platform that combines a live connectome runtime, session-level analysis, controlled perturbation testing, operational simulation lanes, agent R&D protocols, and baseline-vs-brain-powered benchmark experiments. This paper presents NeuroAI as a synthetic cognitive laboratory: a system that exposes internal state, records brain-inspired dynamics, runs controlled receptor-level probes, and compares task behavior across experimental conditions.

The attached evidence set includes a Human Connectome session report for **session_1781227624519_kpt65**, an Agent R&D training report, a core brain experiment CSV, a prior Triaxion-47 pharmacological probe packet, and the NeuroEmergence Core / Battle Ops Lab live surface. The Human Connectome session recorded 4,807 ticks over 341.0 seconds, normalized Shannon entropy of 0.841, plasticity index of 0.137, and 17 high-confidence thought events. The core brain experiment compared 30 baseline runs against 30 brain-powered runs for **threat-memory-navigation**, where brain-powered runs achieved mean task success of 0.9881 and route efficiency of 0.9941 while increasing average latency from 4.5255 ms to 16.1677 ms.

The central contribution is not a biological or clinical claim. The central contribution is a runtime architecture for synthetic cognition research: NeuroAI can

generate structured evidence about internal state, perturbation response, adaptive behavior, and operational training while preserving public safety boundaries around what the results do and do not prove.

1. Introduction

Artificial intelligence systems are usually evaluated from the outside: prompt in, answer out, score the result. That model is not enough for synthetic cognition. A cognitive runtime must expose more than final outputs. It must expose state, tension, adaptation, control, memory, perturbation response, and failure boundaries.

NeuroAI is built around that principle. It treats intelligence as a measurable runtime rather than a single text-generating surface. The platform joins a synthetic connectome, session reports, pharmacological perturbation probes, R&D training lanes, Battle Ops simulation, and benchmark experiments into one research substrate.

This paper asks a simple question:

What does it mean for an AI system to become inspectable as a synthetic cognitive organism without pretending it is a biological brain?

The answer is a bounded research architecture. NeuroAI is not presented here as a medical device, a conscious biological organism, or a deployed operational system. It is presented as a synthetic cognitive laboratory: a build-derived runtime where cognition-like behavior can be simulated, measured, stressed, compared, and packaged into evidence.

2. System Overview

NeuroAI has six visible research surfaces.

Surface	Function	Evidence
NeuroEmergence Core	live synthetic connectome, activity/weight view, cognitive governance, system monitoring	runtime screenshots and session report
Human Connectome Reports	entropy, plasticity, saturation warnings, thought logs, behavioral events	<code>session_1781227624519_kpt65</code> report

Surface	Function	Evidence
Pharmacology Lab	receptor-targeted controlled synthetic-brain perturbations	Triaxion-47 probe packet
Battle Ops Lab	operational simulation and stress-navigation surface	live NeuroEmergence Core URL
Agent R&D	training lanes for reasoning, resilience, experiment design, and market modeling	R&D report
Core Brain Experiments	baseline vs brain-powered benchmarking	<code>core_brain_exp_1781225128938_2.csv</code>

These surfaces matter because they form a complete observation loop: internal state, perturbation, behavior, training, and comparison.

3. Session-Level Connectome Evidence

The Human Connectome session report recorded:

Metric	Value
Session ID	<code>session_1781227624519_kpt65</code>
Duration	341.0 seconds
Ticks	4,807
Estimated tick interval	~70.9 ms
Normalized Shannon entropy	0.841
Plasticity index	0.137
High-confidence thoughts	17

The report uses careful claim language: “emergent behavior in a brain-inspired embodied simulation.” It also includes a saturation warning: five regions averaged above 90% activation. This warning is important. It means the instrumentation is not merely promotional; it is capable of reporting validation risk.

The strongest release-safe interpretation is:

NeuroAI produced a session-level synthetic connectome report with entropy, plasticity, high-confidence thought events, behavioral events, and a saturation warning requiring homeostatic-scaling review before stronger emergence claims.

4. Controlled Perturbation Evidence

The Triaxion-47 packet records a controlled synthetic pharmacology probe. The run tested a synthetic tri-mix antipsychotic candidate with a 50 mg equivalent dose and BBB coefficient of 0.30 against D1, D2, NMDA, AMPA, GABA-A, and 5-HT1A receptor targets.

Key result:

- peak receptor occupancy: 2.2%
- regional activation deltas: -8.5% to -8.7%
- integrity: PASS
- stability: PASS
- verdict: WEAK EFFECT
- ANSInterceptiveCoupling: no measurable separation from baseline or shuffled control

The significance is methodological. NeuroAI can accept receptor-level perturbation inputs, propagate them through region-mapped synthetic modules, and return a bounded stability verdict. This supports a protocol candidate for synthetic pharmacology stress testing. It does not support human treatment, dosing, efficacy, or biological significance claims.

5. Agent R&D Evidence

The Agent R&D report records four training lanes.

Lane	Protocol	Readiness	Validation
AI Intelligence	Reasoning Depth	96%	Weak effect
Military / Political	Stress Resilience	96%	Weak effect
Research	Autonomous Experiment Design	76%	Weak effect
Commercialization	Market Pattern Modeling	90%	Weak effect

The public-safe reading is that NeuroAI supports protocolized training lanes with readiness and divergence signals. The stress-resilience lane should be described as simulation and operational decision training, not as deployed political or military capability.

6. Brain-Powered Benchmark Evidence

The core brain experiment compared 30 baseline runs with 30 brain-powered runs in the `threat-memory-navigation` scenario.

Metric	Baseline	Brain-Powered	Delta
Task success	0.0000	0.9881	+0.9881
Route efficiency	0.0000	0.9941	+0.9941

Metric	Baseline	Brain-Powered	Delta
Adaptation rate	0.0875	0.9875	+0.9000
Recovery success	0.2375	0.8438	+0.6063
Threat avoidance	0.0075	0.5788	+0.5713
Coherence score	0.1875	1.0000	+0.8125
Emergence score	0.1000	1.0000	+0.9000
Novelty score	0.0000	1.0000	+1.0000
Thought diversity	0.1500	0.8667	+0.7167
Useful emergence count	0.0000	9.8667	+9.8667
Autonomic balance stability	0.4212	0.7459	+0.3247
Stress magnitude	0.5517	0.3535	-0.1981
Recovery slope	0.2688	0.7163	+0.4475
Self-state coherence	0.3262	0.7460	+0.4198
Average latency	4.5255 ms	16.1677 ms	+11.6422 ms

The benchmark result is strong and interpretable: brain-powered runs improved success, adaptation, recovery, coherence, emergence, novelty, and self-state coherence, while increasing latency. This is a real engineering tradeoff. NeuroAI buys richer cognition-like processing at a measurable compute/time cost.

7. Battle Ops Lab

The Battle Ops Lab is part of the NeuroEmergence Core live surface. In this release, it should be framed as an operational simulation lane for stress, threat-memory-navigation, strategy-shift, resilience, and decision-pressure experiments.

Public-safe wording:

Battle Ops Lab is an operational simulation surface inside NeuroAI for studying decision pressure, threat-memory-navigation, stress resilience, and strategy adaptation in a synthetic cognitive runtime.

Blocked wording:

Battle Ops Lab is a deployed military command system.

The distinction is non-negotiable for release safety.

8. AI Psychology Thesis

NeuroAI suggests a concrete research program for AI psychology. Instead of treating AI psychology as personality labels or chat behavior, NeuroAI treats it as runtime state under pressure.

An AI psychology stack should measure:

- internal state coherence

- stress magnitude
- recovery slope
- autonomic balance stability
- thought diversity
- useful emergence
- latency under cognitive load
- response to perturbation
- saturation and homeostatic recovery

This is the deeper theorem:

AI psychology becomes scientifically tractable when mental-state language is converted into measurable runtime variables, perturbation responses, recovery curves, and evidence-bounded session reports.

That theorem is a protocol candidate. The current evidence set makes it credible enough for a Zenodo working paper, with stronger validation requiring repeated sessions, ablation runs, and public redacted receipts.

9. Claim Boundary

Claim	Public Status	Evidence
NeuroAI is a build-derived synthetic cognition runtime	Supported internal result	attached reports, screenshots, live surface
NeuroAI records connectome session metrics	Supported internal result	Human Connectome session PDF
NeuroAI supports controlled perturbation probes	Supported internal result / protocol candidate	Triaxion-47 packet
Brain-powered runs outperform baseline in the uploaded CSV	Supported internal result	experiment CSV
Battle Ops Lab is an operational simulation lane	Supported internal result / protocol candidate	operator statement and live surface
NeuroAI is biologically conscious	Blocked	not supported
NeuroAI diagnoses or treats humans	Blocked	not supported
Synthetic pharmacology replaces biological validation	Blocked	not supported

Claim	Public Status	Evidence
Battle Ops is deployed operational capability	Blocked	not supported

10. Zenodo Release Package

The Zenodo-ready package should include:

- paper PDF
- paper DOCX
- paper markdown
- journal-style HTML page
- Zenodo metadata JSON
- Citation CFF
- claims matrix
- evidence matrix
- release boundary
- hash manifest
- main Medina/THESIS logo
- NeuroAI topic logo
- supplemental source artifacts where release-approved

11. Conclusion

NeuroAI is ready for a serious public working paper because it now has more than a claim. It has a system map, attached session report, pharmacological probe record, agent training report, benchmark CSV, and a live operational surface. The strongest public contribution is not that NeuroAI is a biological brain. The strongest contribution is that NeuroAI provides a measurable synthetic cognitive laboratory where internal state, perturbation response, adaptive behavior, and operational training can be packaged as evidence.

That is the beginning of a publishable AI psychology program.

References

1. NeuroEmergence Core live surface. <https://fond-pink-nvy-draft.caffeine.xyz/>
2. Allen Brain Atlas. <https://atlas.brain-map.org/>
3. Human Connectome Project. <https://www.humanconnectome.org/>
4. Glasser et al., “A multi-modal parcellation of human cerebral cortex.” <https://pmc.ncbi.nlm.nih.gov/articles/PMC4990127/>
5. Brainnetome Atlas. <https://atlas.brainnetome.org/>
6. Fan et al., “The Human Brainnetome Atlas.” <https://pmc.ncbi.nlm.nih.gov/articles/PMC4961028/>
7. Zenodo Developers Documentation. <https://developers.zenodo.org/>
8. Zenodo help, “Describe records.” <https://help.zenodo.org/docs/deposit/describe-records/>